

ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

- **English Name:** Grace (HSU Chia-Hsin)
 - **Chinese Name:** 許家忻
 - **Email:** gracee1109@gapp.nthu.edu.tw
 - **Student ID:** 111096033
 - **University:** National Tsing Hua University
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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
 - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
 - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.
- **Prohibited (AI as Solver):**
 - Copying assignment questions directly into an AI tool to generate answers.
 - Using AI to fabricate experimental results or observations without executing the code in [DM2026-Lab1-Master.ipynb](#).
 - Submitting machine-generated prose wholesale.
- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).
- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).
- **When AI was not used:** You may write “No” under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** (*required when AI = Yes; encouraged otherwise*)

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

There are 594 documents in the sci.med category. Isolating specific categories is critical because mixing disparate categories introduces cross-domain noise and causes pattern mining to discover trivial rather than domain-specific associations.

AI used for this answer: Yes

If yes, how: grammar polishing

External references (links): https://en.wikipedia.org/wiki/Association_rule_learning

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

Dropping 40% of the dataset with **dropna()** throws away too much data and can easily mess up the class balance. Since the text itself is still useful, a better alternative is using an unsupervised method like K-Means clustering to group the documents without needing the category labels.

AI used for this answer: Yes

If yes, how: grammar polishing

External references (links):

https://kenndanielso.github.io/mlrefined/blog_posts/11_Linear_unsupervised_learning/1_5_K_means_clustering.html

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses `X_sample = X.sample(n=1000)` to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1, No, the sampled proportions don't match the original perfectly due to random sampling variance, even though they stay within an acceptable range.

2, The main risk is that simple random sampling can easily underrepresent or completely miss minority classes, making the sampled dataset imbalanced and hiding rare but important patterns.

AI used for this answer: No

If yes, how:

External references (links):

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the `CountVectorizer` converts text into a document-term matrix using `stop_words='english'`. Change this parameter to `stop_words=None` and re-run the cell to check the shape of `X_counts`.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1, The total number of features increases because all the common words that were previously filtered out are included in the vocabulary.

2, Adding stopwords back slightly decreases the overall sparsity of the matrix, because these words appear frequently across almost every single document, creating many non-zero entries.

AI used for this answer: Yes

If yes, how: Concept explanation

External references (links): https://scikit-learn.org/stable/modules/generated/sklearn.feature_selection.VarianceThreshold.html

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run `variance_filter` using threshold values `threshold = 0.0`, `threshold = 0.01`, and `threshold = 0.05`. Record the number of features marked as `keep == True`.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold Number of terms kept (keep == True)

0.0=35482

0.01=5089

0.05=1491

1, As the threshold increases from 0.0 to 0.05, the vocabulary size drops from 35,482 down to 1,491 terms because stricter variance cutoffs filter out most low-frequency words.

2, The words eliminated first are primarily extremely rare words that appear only once or twice across the dataset, giving them an almost constant zero count across document

3, Dropping near-zero variance features is safe because words that show almost no variation carry little to no discriminative information between categories, so removing them cuts noise without hurting model accuracy.

AI used for this answer: Yes

If yes, how: phrasing polish, Concept explanation

External references (links): https://en.wikipedia.org/wiki/Feature_selection

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the `comp.graphics` class. Inspect the top 10 features sorted by `pearson_r` and compare them to the top 10 features sorted by `spearman_r`.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?

2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

- 1, The top-correlated words are mostly domain-specific terminology for graphics rather than generic stop words.
- 2, Pearson measures linear relationships between raw counts, while Spearman measures monotonic relationships based on ranked values.
- 3, It happens when a word reliably signals the topic but has skewed counts or extreme outliers in a few texts, which messes up Pearson's linear fit while Spearman's rank order stays solid.

AI used for this answer: Yes

If yes, how: grammar polishing, statistical concept explanation

External references (links):

https://www.researchgate.net/publication/337559994_A_Measure_of_Similarity_in_Textual_Data_Using_Spearman%27s_Rank_Correlation_Coefficient

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100** and **k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9** and **minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter **k** in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

Decreasing minSup from 9 to 3 increase the number of discovered patterns from 710 up to 9,547 because a much wider set of lower-frequency itemset now qualifies. Consequently, the execution runtime increases significantly from around 0.04 seconds to over 0.62 seconds due to the exponential expansion of the FP-tree branches and recursive search space.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

The parameter *k* in Top-K FAE sets an explicit quota on the total number of desired patterns, raising internal thresholds to return only the *k* most frequent one without

needing prior knowledge of the dataset. In contrast, minSup in MaxFPGrowth defines a rigid frequency cutoff where any pattern meeting the absolute support count is retained, meaning the final number of output patterns is variable and unpredictable.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent patterns are more concise because they only retain frequent item sets that have no frequent supersets. By omitting all redundant frequent subsets which standard frequent pattern mining exhaustively enumerates maximal patterns provide a compact, loss-free representation of the itemset boundaries.

AI used for this answer: Yes

If yes, how: algorithmic comparison and phrasing polish

External references (links): <https://pami-dminer.github.io/PAMI/>

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1, A scatter plot can only visualize 2D or 3D coordinates, whereas the raw X_counts matrix resides in a space with tens of thousands of features, making direct geometric plotting mathematically impossible.

2, Dimensionality reduction algorithms must be applied first, specifically techniques such as PCA, t-SNE, or UMAP .

AI used for this answer: No

If yes, how:

External references (links):

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method (**PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1, t-SNE and UMAP show distinct, well-separated cluster islands for the 4 document categories, whereas PCA shows heavy overlap around the center where categories blend.

2, PCA is a linear projection technique that seeks directions maximizing global variance, making it incapable of capturing the non-linear manifold structures and high-dimensional semantic relationships inherent in sparse text data. In contrast, t-SNE and UMAP optimize for preserving local neighborhood structure.

3, t-SNE preserves local similarities rather than global distances. The physical spacing and relative positioning between separate clusters are heavily influenced by arbitrary initialization and hyperparameters, so inter-cluster distances do not reflect true semantic dissimilarity in the original high-dimensional space.

AI used for this answer: Yes

If yes, how: Theoretical comparison and phrasing polish

External references (links): <https://distill.pub/2016/misread-tsne/>

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column (**0, 1, 2, 3**) as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1, Exactly 4 binary columns are created, where each category is represented by an indicator vector with a 1 in its respective column and 0s elsewhere.

2, The algorithm falsely assumes an inherent order and numerical magnitude among the categories assuming $3 > 2 > 1 > 0$ and that category 2 is twice as far from category 0 as category 1 is, even though the classes are nominal and unordered.

3, One-hot binarization maps each class to an independent, mutually orthogonal dimension. This ensures that the geometric distance between any two distinct categories is identical, completely removing the artificial ordering and unequal distance assumptions.

AI used for this answer: Yes

If yes, how: Mathematical concept explanation and phrasing polish

External references (links): <https://machinelearningmastery.com/why-one-hot-encode-data-in-machine-learning/>
